Comprehensive Learning Pathway for Building a Malware Detection System

This pathway includes selected video resources for each section to provide visual learning alongside text-based materials. The learning plan balances theory with hands-on practice and creates a clear progression from foundational skills to an advanced malware detection product.

PHASE 1: Core Foundations (0-2 months)

1. Programming with Rust (3 weeks)

- **Week 1:** Rust fundamentals
  - The Rust Book (Official) - Chapters 1-9
  - **Video:** Rust Crash Course by Traversy Media (1hr 40min)
  - Rustlings exercises for practice
- **Week 2:** Advanced Rust concepts
  - The Rust Book - Chapters 10-20
  - **Video:** Crust of Rust series by Jon Gjengset
  - Focus on ownership, lifetimes, concurrency, and error handling
- **Week 3:** Practical applications
  - Rust By Example
  - **Video:** Systems Programming with Rust by Ryan Levick (2hrs)
  - Complete 15-20 exercises on Exercism's Rust track

2. Linux & Operating System Concepts (3 weeks)

- **Week 1:** Linux basics
  - Linux Journey - Complete "Grasshopper" and "Journeyman" sections
  - **Video:** Linux Tutorial for Beginners by freeCodeCamp (6hrs)
  - Master commands: ls, cd, cat, grep, chmod, ps, top, find, netstat
- **Week 2:** System monitoring and analysis
  - Learn tools: strace, lsof, tcpdump, wireshark
  - **Video:** Linux Command Line Tools Every Developer Should Know by gotbletu
- **Week 3:** Windows OS internals
  - Windows Internals book (at least parts 1-3)

- **Video:** Windows Internals by Pavel Yosifovich (1hr 30min)
- Process Explorer, Process Monitor, and Windows API basics

3. Version Control & Project Management (2 weeks)

- **Week 1:** Git fundamentals
  - Pro Git Book (Chapters 1-5)
  - **Video:** Git and GitHub for Beginners by freeCodeCamp (1hr 10min)
  - Practice with GitHub by creating your own repository
- **Week 2:** Advanced Git and CI/CD
  - GitHub Actions for automated testing
  - **Video:** GitHub Actions Tutorial by TechWorld with Nana (1hr)
  - Branch protection and code review workflows
  - GitHub Flow methodology

PHASE 2: Malware Analysis Fundamentals (2-5 months)

4. File Formats & Binary Analysis (4 weeks)

- **Week 1-2:** Windows PE files
  - PE format specification
  - **Video:** PE Format & Malware Analysis by MalwareAnalysisForHedgehogs (40min)
  - Rust crate `goblin` for parsing PE files
  - Practice extracting headers, sections, imports/exports
- **Week 3:** ELF files (Linux)
  - ELF format specification
  - **Video:** ELF Files - Understanding the Executable and Linkable Format by LiveOverflow (20min)
  - Analyze samples with `readelf` and Rust's `goblin` crate
- **Week 4:** Memory mapping and binary parsing
  - Using `memmap2` in Rust
  - **Video:** Binary Analysis with Rust by LiveOverflow (30min)
  - Building a basic file analyzer that extracts metadata from binaries

5. Static Analysis Techniques (3 weeks)

- **Week 1:** Disassembly concepts

- Learn assembly basics (x86/x64)
- **Video:** Introduction to Reverse Engineering with Ghidra by freeCodeCamp (4hrs)
- Use `capstone` crate for disassembly in Rust

- **Week 2:** String extraction and analysis
  - Extract and analyze strings from binaries
  - **Video:** Malware Static Analysis by 13Cubed (36min)
  - Identify suspicious patterns and indicators
- **Week 3:** Advanced static analysis
  - Control flow graph analysis
  - Entropy calculation for detecting packed/encrypted code
  - **Video:** Advanced Malware Analysis by SANS Institute (1hr)
  - Practical Malware Analysis book (Chapters 1-7)

6. Dynamic Analysis & Sandboxing (3 weeks)

- **Week 1:** Basic dynamic analysis
  - Process monitoring techniques
  - API call hooking concepts
  - **Video:** Malware Analysis Tutorials by OALabs
- **Week 2:** Sandbox environment setup
  - Create isolated VM environment for analysis
  - **Video:** Building a Malware Analysis Lab by The PC Security Channel (15min)
  - Study Cuckoo Sandbox architecture
- **Week 3:** Automated behavior analysis
  - Track file system, registry, and network activity
  - **Video:** Dynamic Malware Analysis by John Hammond (1hr 15min)
  - Build a basic behavior analyzer in Rust
  - Zero2Automated Malware Analysis course

7. Yara Rules & Signature-Based Detection (2 weeks)

- **Week 1:** Yara basics
  - Yara documentation
  - **Video:** Getting Started with Yara by SANS (1hr)

- Write basic signature rules
- Integrate with your Rust analyzer via FFI or the `yara` crate

- **Week 2:** Advanced Yara techniques
  - Complex condition logic
  - Memory scanning with Yara
  - **Video:** Advanced YARA by CyberDefenders (1hr 15min)
  - Building a rule repository for your detector

PHASE 3: Machine Learning for Threat Detection (5-8 months)

8. Python & Data Science Fundamentals (3 weeks)

- **Week 1:** Python refresher and environment setup
  - Set up Python with proper virtual environments
  - **Video:** Python Tutorial for Beginners by Programming with Mosh (6hrs)
  - Jupyter notebooks for exploration
- **Week 2-3:** Data science libraries
  - `numpy` and `pandas` for data manipulation
  - `matplotlib` and `seaborn` for visualization
  - **Video:** Python for Data Science by freeCodeCamp (4hrs)

9. Feature Engineering for Malware Detection (3 weeks)

- **Week 1:** Static features
  - Extract header information, section properties
  - **Video:** Feature Extraction for Machine Learning by Krish Naik (30min)
  - Calculate entropy, size ratios, import counts
- **Week 2:** Dynamic features
  - API call sequences
  - Network communication patterns
  - **Video:** Machine Learning for Cybersecurity by SANS Institute (45min)
  - System changes (files, registry)
- **Week 3:** Create feature extraction pipeline
  - Build an end-to-end pipeline in Python
  - **Video:** Building ML Pipelines with Scikit-learn by DataCamp (1hr)

- Create labeled dataset from known malware/benign samples

10. ML Model Development (4 weeks)

- **Week 1:** Model selection and training
  - Try `RandomForest`, `XGBoost`, `LightGBM`
  - **Video:** ML Algorithms Explained by StatQuest
  - Cross-validation techniques
- **Week 2:** Model evaluation and improvement
  - Precision/recall metrics
  - ROC curves and AUC analysis
  - **Video:** Evaluation Metrics for Machine Learning by Krish Naik (45min)
  - Scikit-learn documentation
- **Week 3:** Model optimization
  - Hyperparameter tuning
  - Feature importance analysis
  - **Video:** Hyperparameter Tuning in Python by Kaggle (40min)
- **Week 4:** ML model deployment
  - Serialize models with `joblib`
  - Create REST API with FastAPI for model serving
  - **Video:** FastAPI Tutorial by Tech With Tim (1hr)
  - Plan for integration with Rust backend

11. Threat Intelligence Integration (2 weeks)

- **Week 1:** Threat intelligence concepts
  - IOCs (Indicators of Compromise)
  - MITRE ATT&CK framework
  - **Video:** Threat Intelligence Fundamentals by SANS (1hr)
- **Week 2:** Integrate threat feeds
  - Connect to AlienVault OTX, VirusTotal API
  - **Video:** Threat Intelligence Automation by Black Hills InfoSec (1hr)
  - Build a threat intelligence database
  - MITRE ATT&CK framework

PHASE 4: Building the Complete System (8-10 months)

12. Rust Backend Development (4 weeks)

- **Week 1-2:** Web framework fundamentals
  - Learn `actix-web` or `axum`
  - **Video:** Rust Web Development with Actix by Let's Get Rusty (2hrs)
  - Set up API routes for file scanning
- **Week 3-4:** Full backend implementation
  - File upload handling
  - Scanner orchestration
  - Database integration (PostgreSQL with `sqlx` crate)
  - **Video:** Rust Web API Full Course by Brooks Builds
  - Zero To Production In Rust book

13. Database Design & Implementation (2 weeks)

- **Week 1:** Database schema design
  - Plan tables for scans, results, users, threat intel
  - Learn normalization principles
  - **Video:** Database Design Course by freeCodeCamp (8hrs)
- **Week 2:** Database implementation
  - Set up PostgreSQL
  - **Video:** PostgreSQL Tutorial by Amigoscode (2hrs)
  - Implement migrations and connection pooling
  - Build repository pattern for data access

14. Frontend Development (4 weeks)

- **Week 1:** React fundamentals
  - Set up React with TypeScript
  - **Video:** React Course for Beginners by freeCodeCamp (11hrs)
- **Week 2:** Component design
  - Create file upload components
  - Design dashboard and reports layout

    - **Video:** React Hooks Tutorial by Net Ninja
- **Week 3-4:** Complete UI implementation
    - Connect to backend API
    - **Video:** Building a Full-Stack App with React by Traversy Media (2hrs)
    - Implement visualization of scan results
    - Add user authentication flow
    - **Video:** Material UI Tutorial by Code Evolution
    - Material UI or Chakra UI for components

15. WebAssembly Integration (Optional, 2 weeks)

- **Week 1:** WASM fundamentals
    - Learn how to compile Rust to WASM
    - **Video:** WebAssembly Crash Course by Traversy Media (1hr)
    - Rust and WebAssembly book
- **Week 2:** In-browser scanner implementation
    - Compile core scanner functionality to WASM
    - **Video:** Rust and WebAssembly Tutorial by Brooks Builds
    - Integrate with React frontend

PHASE 5: AI Enhancements & Production Readiness (10-12 months)

16. AI Report Generation (3 weeks)

- **Week 1:** LLM API integration
    - Set up OpenAI or Gemini API access
    - **Video:** OpenAI API Crash Course by Adrian Twarog (1hr 15min)
    - Design prompt templates for analysis reports
- **Week 2-3:** Report generation system
    - Create structured data input for AI
    - **Video:** Building AI Applications with LangChain by Patrick Loeber (1hr 30min)
    - Implement context-aware report generation
    - Add feedback loop for improving reports

17. Security Hardening (2 weeks)

- **Week 1:** Security audit
  - Code review for vulnerabilities
  - Input validation and sanitization
  - **Video:** Web Security Fundamentals by Academind (1hr 30min)
  - Proper error handling
- **Week 2:** Penetration testing
  - Fuzzing the file analyzer
  - API security testing
  - **Video:** Bug Bounty Hunting Methodology by Nahamsec (2hrs)
  - Authentication system review
  - OWASP Top 10

18. Deployment & DevOps (3 weeks)

- **Week 1:** Containerization
  - Create Docker containers for all components
  - **Video:** Docker Crash Course by TechWorld with Nana (3hrs)
  - Compose multi-container application
- **Week 2:** CI/CD pipeline
  - Set up GitHub Actions for testing and deployment
  - **Video:** Complete CI/CD Pipeline by DevOps Journey (1hr)
  - Implement automated security scanning
- **Week 3:** Cloud deployment
  - Deploy to AWS, GCP, or Azure
  - **Video:** AWS for Beginners by freeCodeCamp (4hrs)
  - Set up monitoring and logging
  - Implement backup strategies

PHASE 6: Advanced Features & Growth (Beyond 12 months)

19. Advanced Detection Techniques

- Behavior-based detection improvements
- Memory forensics integration
- **Video:** Advanced Memory Forensics by SANS (1hr)

- Network traffic analysis
- SANS FOR610: Reverse-Engineering Malware

20. Performance Optimization

- Profiling and benchmarking
- **Video:** Performance Optimization for Web Applications by Google Chrome Developers (1hr)
- Distributed scanning architecture
- Caching strategies

21. Productization & Growth

- User feedback integration
- Feature prioritization
- **Video:** Building Products People Want by Y Combinator (40min)
- Threat intelligence sharing capabilities

Learning Resources Summary

Books

- The Rust Book (Official)
- Practical Malware Analysis (Michael Sikorski & Andrew Honig)
- Windows Internals (Pavel Yosifovich et al.)
- Zero To Production In Rust (Luca Palmieri)

Online Courses

- Rust by Example
- Zero2Automated Malware Analysis
- freeCodeCamp React Course
- SANS FOR610

Platforms for Practice

- Exercism (Rust)
- Any.Run (for malware analysis)
- MalwareBazaar (for samples)
- HackTheBox (security challenges)

YouTube Channels to Follow

- John Hammond - Malware analysis and security
- LiveOverflow - Binary exploitation and security
- OALabs - Malware reverse engineering
- 13Cubed - Digital forensics and incident response
- Let's Get Rusty - Rust programming
- freeCodeCamp - General programming tutorials
- SANS Offensive Operations - Security techniques